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```
clc;
clear;
close all;
```

===== PARAMETERS

=====

```
M = 16;                % 16-QAM
k = log2(M);

Nfft = 64;             % Number of OFDM subcarriers
Nofdm = 100;           % Number of OFDM symbols
CP = 16;               % Cyclic Prefix

SNR_dB = 0:5:25;

BER_AWGN = zeros(size(SNR_dB));
BER_NoEQ = zeros(size(SNR_dB));
BER_EQ = zeros(size(SNR_dB));
```

===== TRANSMITTER

=====

```
% Generate random bits
Nbits = Nofdm * Nfft * k;

txBits = randi([0 1], Nbits, 1);

% 16-QAM modulation
```

```
txSymbols = qammod(txBits, M, ...
    'InputType', 'bit', ...
    'UnitAveragePower', true);

% Arrange symbols into OFDM symbols
txSymbolsMatrix = reshape(txSymbols, Nfft, Nofdm);

% IFFT
txOFDM = ifft(txSymbolsMatrix, Nfft);
```

===== CYCLIC PREFIX

```
=====
```

```
% Add cyclic prefix
cyclicPrefix = txOFDM(end-CP+1:end, :);

txOFDM_CP = [cyclicPrefix; txOFDM];
```

===== MULTIPATH CHANNEL

```
=====
```

```
% Three-path channel
h = [1 0.5 0.3];

% Channel frequency response
H = fft([h zeros(1, Nfft-length(h))], Nfft).';
```

===== SNR LOOP

```
=====
```

```
for i = 1:length(SNR_dB)
```

```
-----
```

CASE 1: AWGN ONLY

```
-----
```

```
    rxAWGN = awgn(txOFDM, SNR_dB(i), 'measured');

    % FFT
    rxSymbols_AWGN = fft(rxAWGN, Nfft);

    % Convert to serial
    rxSymbols_AWGN = rxSymbols_AWGN(:);

    % Demodulation
```

```
rxBits_AWGN = qamdemod(rxSymbols_AWGN, M, ...  
    'OutputType', 'bit', ...  
    'UnitAveragePower', true);
```

```
% BER  
[~, BER_AWGN(i)] = biterr(txBits, rxBits_AWGN);
```

CASE 2: MULTIPATH + AWGN WITHOUT EQUALIZATION

```
rxMultipath = zeros(size(txOFDM_CP));  
  
for n = 1:Nofdm  
    rxMultipath(:,n) = filter(h, 1, txOFDM_CP(:,n));  
  
end  
  
% Add AWGN  
rxNoEQ = awgn(rxMultipath, SNR_dB(i), 'measured');  
  
% Remove CP  
rxNoEQ = rxNoEQ(CP+1:end, :);  
  
% FFT  
rxSymbols_NoEQ = fft(rxNoEQ, Nfft);  
  
% Serial conversion  
rxSymbols_NoEQ = rxSymbols_NoEQ(:);  
  
% Demodulation  
rxBits_NoEQ = qamdemod(rxSymbols_NoEQ, M, ...  
    'OutputType', 'bit', ...  
    'UnitAveragePower', true);  
  
% BER  
[~, BER_NoEQ(i)] = biterr(txBits, rxBits_NoEQ);
```

CASE 3: MULTIPATH + AWGN WITH EQUALIZATION

```
% FFT  
rxSymbols_EQ = fft(rxNoEQ, Nfft);  
  
% Equalization
```

```

rxSymbols_EQ = rxSymbols_EQ ./ H;

% Serial conversion
rxSymbols_EQ = rxSymbols_EQ(:);

% Demodulation
rxBits_EQ = qamdemod(rxSymbols_EQ, M, ...
    'OutputType', 'bit', ...
    'UnitAveragePower', true);

% BER
[~, BER_EQ(i)] = biterr(txBits, rxBits_EQ);

```

Display results

```

fprintf(['SNR = %2d dB | ' ...
    'AWGN BER = %.4e | ' ...
    'No EQ BER = %.4e | ' ...
    'EQ BER = %.4e\n'], ...
    SNR_dB(i), ...
    BER_AWGN(i), ...
    BER_NoEQ(i), ...
    BER_EQ(i));

```

SNR = 0 dB | AWGN BER = 2.8805e-01 | No EQ BER = 3.3633e-01 | EQ BER = 3.1020e-01

SNR = 5 dB | AWGN BER = 1.6543e-01 | No EQ BER = 2.5250e-01 | EQ BER = 1.9801e-01

SNR = 10 dB | AWGN BER = 5.8359e-02 | No EQ BER = 1.9535e-01 | EQ BER = 9.3398e-02

SNR = 15 dB | AWGN BER = 4.6875e-03 | No EQ BER = 1.6469e-01 | EQ BER = 2.6016e-02

SNR = 20 dB | AWGN BER = 0.0000e+00 | No EQ BER = 1.4789e-01 | EQ BER = 9.3750e-04

SNR = 25 dB | AWGN BER = 0.0000e+00 | No EQ BER = 1.3836e-01 | EQ BER = 0.0000e+00

end

===== COMPARISON GRAPH

=====

```

figure;

semilogy(SNR_dB, BER_AWGN, '-o', 'LineWidth', 1.5);
hold on;

semilogy(SNR_dB, BER_NoEQ, '-s', 'LineWidth', 1.5);

```

```

semilogy(SNR_dB, BER_EQ, '-^', 'LineWidth', 1.5);

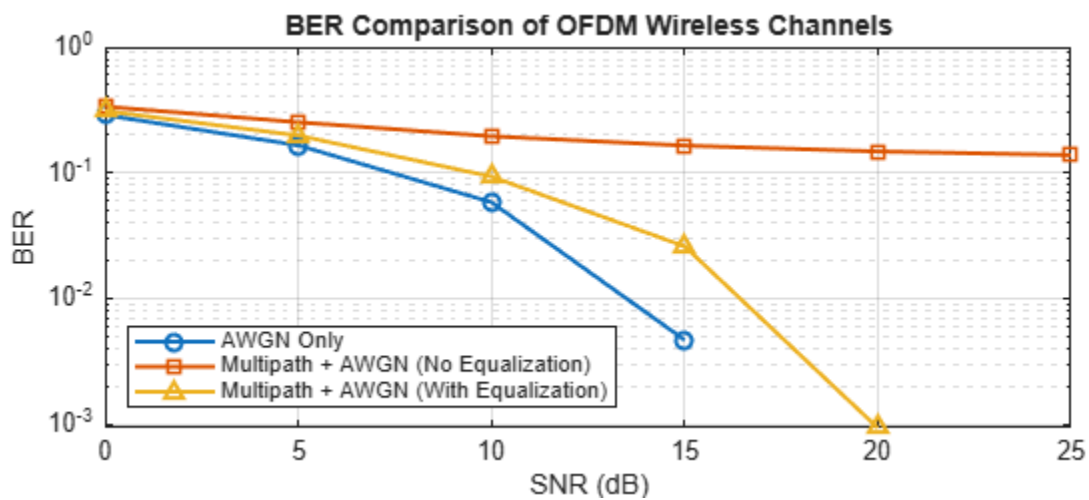
grid on;

xlabel('SNR (dB)');
ylabel('BER');

title('BER Comparison of OFDM Wireless Channels');

legend('AWGN Only', ...
       'Multipath + AWGN (No Equalization)', ...
       'Multipath + AWGN (With Equalization)', ...
       'Location', 'southwest');

```



===== FINAL CONSTELLATION

```

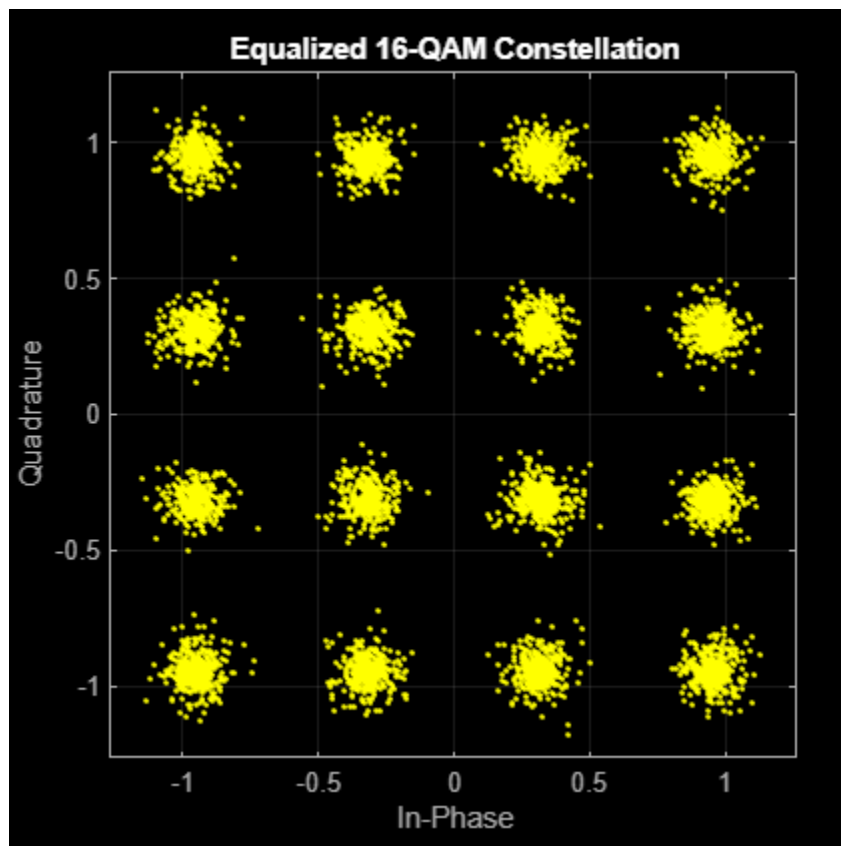
figure;

scatterplot(rxSymbols_EQ);

title('Equalized 16-QAM Constellation');

grid on;

```



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